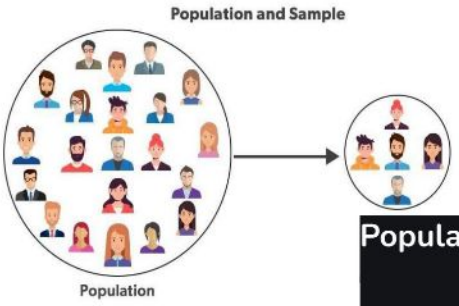
UNIT-IV

STATISTICAL INFERENCE AND ANALYSIS

Population & Sample



- A population refers to the entire set of individuals, objects, or data points that you want to study.
- A sample is a subset of the population that is selected for analysis.

Population vs Sample

Population	Sample
The population includes all members of a specified group.	A sample is a subset of the population.
Collecting data from an entire population can be time-consuming, expensive, and sometimes impractical or impossible.	Samples offer a more feasible approach to studying populations, allowing researchers to draw conclusions based on smaller, manageable datasets
Includes all residents in the city.	Consists of 1000 households, a subset of the entire population.

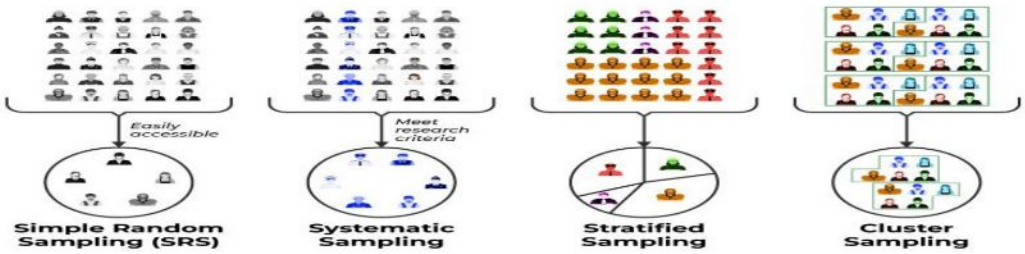
Types of Sampling

THERE ARE 2 TYPES OF SAMPLING:

Non-Probability sampling

Probability sampling

- SIMPLE RANDOM SAMPLING
- SYSTEMATIC SAMPLING
- STRATIFIED SMPLING
- CLUSTER SAMPLING
- MULTISTAGE AREA



Simple Random Sampling

Under this scheme, a list of all the elements of the population from where the samples to be drawn is prepared. These are put in a box and shuffled properly.

Example: Suppose there are 75 students in a class and it is decided to select 15 out of 75 students. These students can be numbered from 01 to 75 and can be selected randomly.

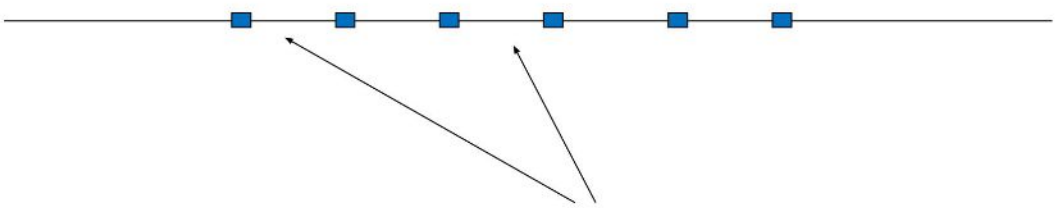


The gap, or period between successive elements is random, uneven, has no particular pattern.

Systematic Sampling

In systematic sampling, the entire population is arranged I a particular order. The order could be calendar dates, alphabetical order which are usually assumed to be random in order. Once this is done, the steps involved in the systematic sampling design are:

(i) Number the units in population from 1 to N
(ii) Decide on the n (sample size) that is required
(iii) Select an interval size k = N/n
(iv) Randomly select an integer between 1 to k
(v) Finally, take every $_{kth}$ unit.



Gaps between elements are equal and Constant□ There is periodicity.

Stratified & Cluster Sampling

Stratified Sampling

Divide the population by certain characteristics into homogeneous subgroups (**strata**) (e.g., UI PhD students, Masters Students, Bachelors students).

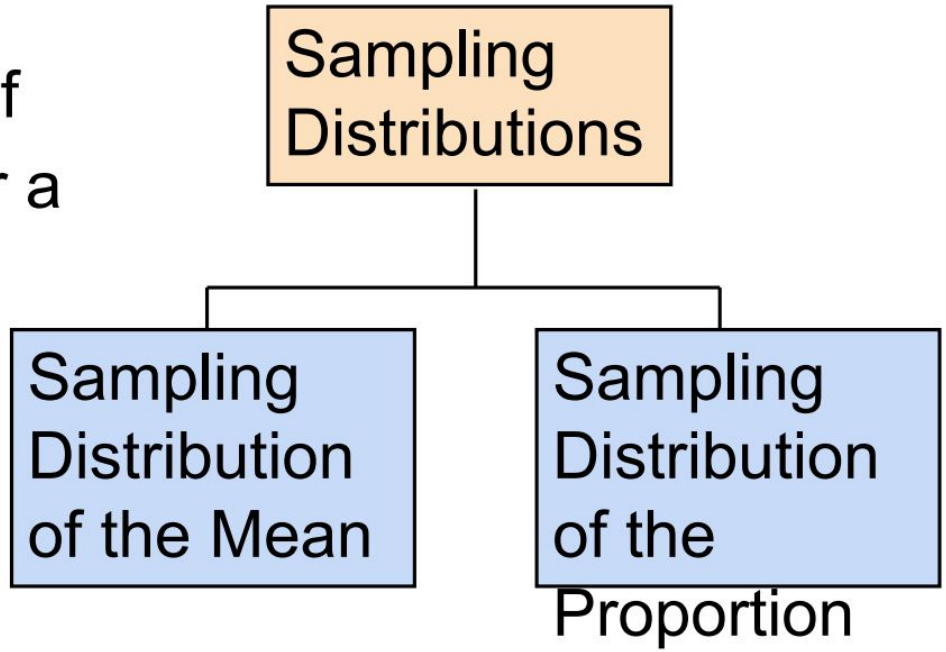
- Elements ***within*** each strata are homogeneous, but are heterogeneous ***across*** strata.

Cluster Sampling

Some populations are spread out (over a state or country).
Elements occur in clumps (towns, districts)—Primary sampling units (PSU).
Elements are hard to reach and identify.
Trade accuracy for efficiency.

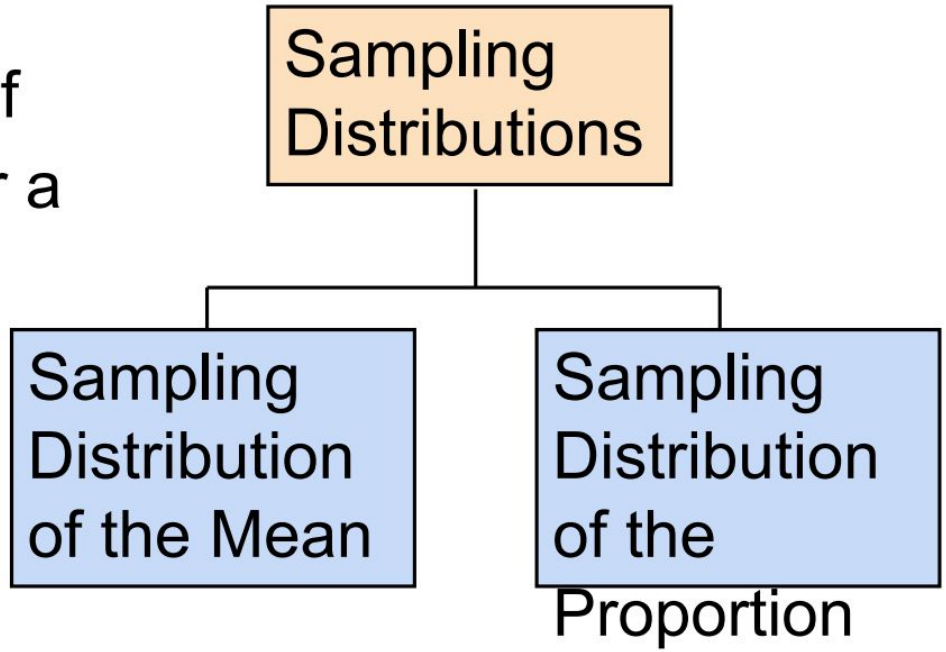
Sampling Distribution

A sampling distribution is a distribution of all of the possible values of a statistic for a given size sample selected from a population



Sampling Distribution

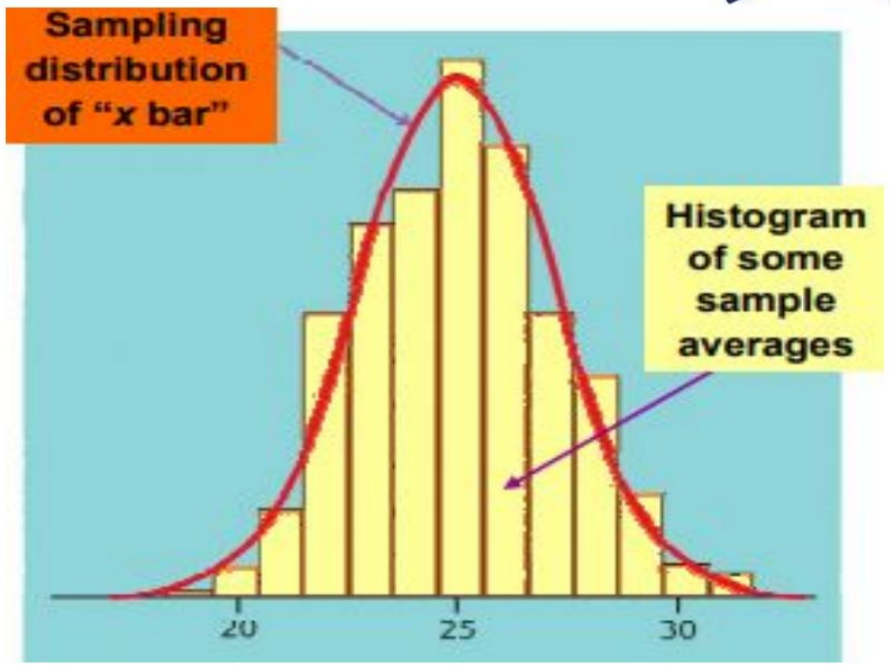
A sampling distribution is a distribution of all of the possible values of a statistic for a given size sample selected from a population



Sampling Distribution of Mean

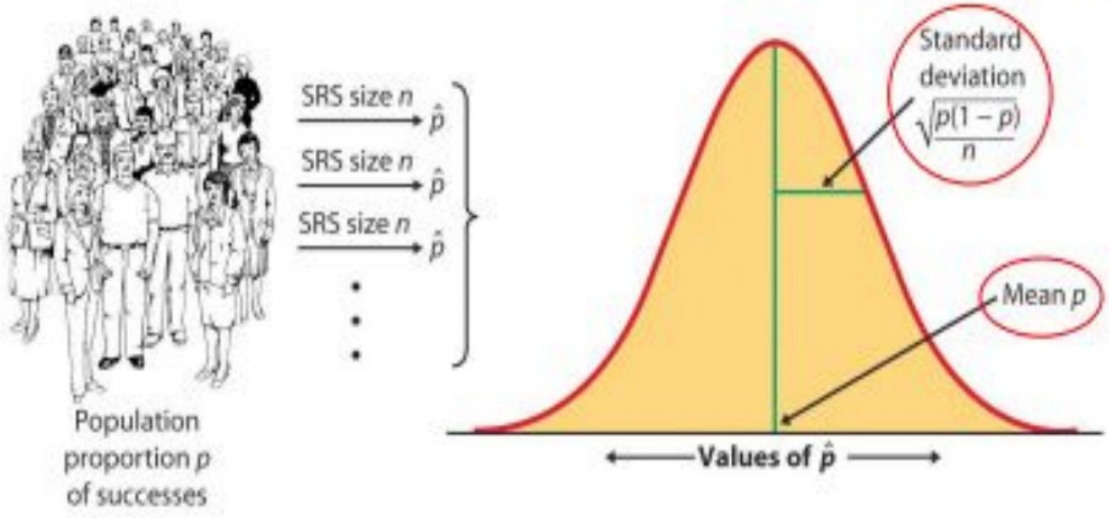
Mean is the most common type of sampling distribution.

It focuses on calculating the mean or rather the average of every sample group chosen from the population and plotting the data points.
The graph shows a normal distribution where the center is the mean of the sampling distribution, which represents the mean of the entire population.



Sampling Distribution of Proportion

- Sampling distribution of proportion focuses on proportions in a population.
- Here, you select samples and calculate their corresponding proportions.
- The means of the sample proportions from each group represent the proportion of the entire population.



Formula for the sampling distribution of a proportion (often denoted as p̂) is:

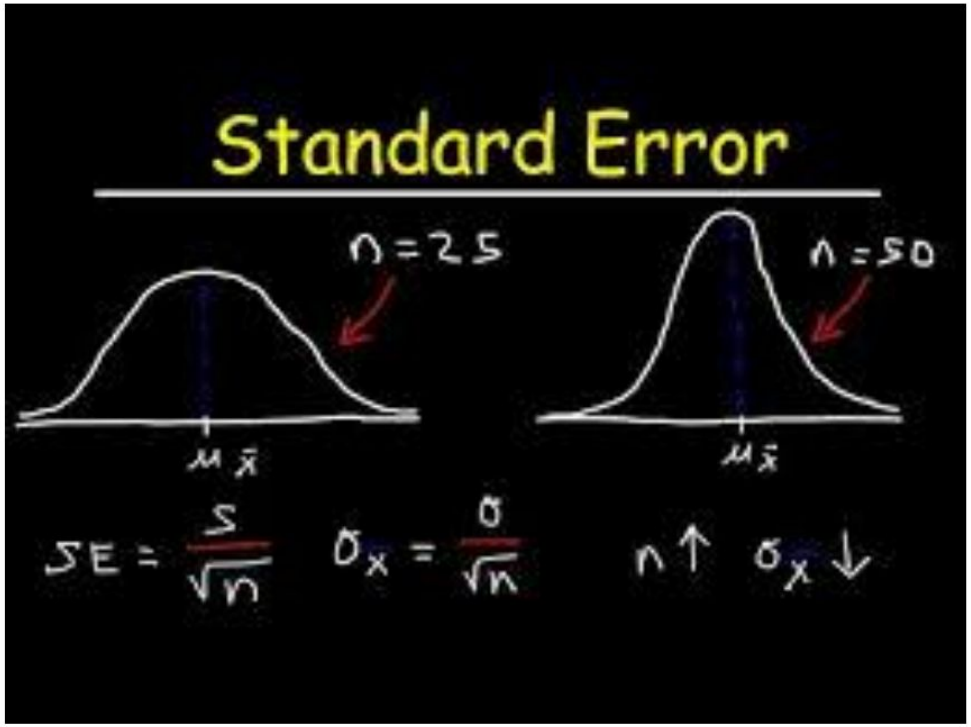
$$\hat{p} = x/n$$

where:

- **p̂** is Sample Proportion
- **x** is Number of "successes" or occurrences of Event of Interest in Sample
- **n** is Sample Size

Standard Error

- Standard error represents how well a given sample represents the population. The Standard Error indicates how well the sample mean predicts the real population mean.
- **Standard Error** is the measure of the variability of a sample statistic used to estimate the variability of a population. Standard Error is important in dealing with sample statistics, such as sample mean, sample proportion, etc.



Standard Error

The Standard Error (SE) formula is used to determine the reliability of a sampling that represents a population. The sample mean that differs from the provided population and is expressed as:

$$SE = S/\sqrt{(n)}$$

where,

- **S** is Standard Deviation of Data
- **n** is Number of Observations

Standard Error of Mean (SEM)

Standard Error of Mean(SEM) is also known by the name Standard Deviation of Mean, is the standard deviation of the measure of sample mean of the population. Standard Error is used when we have to find the population mean from the sample of data and we take two samples to get two different mean in that case we use Standard Error Mean.

Standard Error Mean formula is the ratio of standard deviation to the root of sample size,

$$SEM = S/\sqrt{(n)}$$

where,

- S is Standard Deviation
- n is Number of Observations

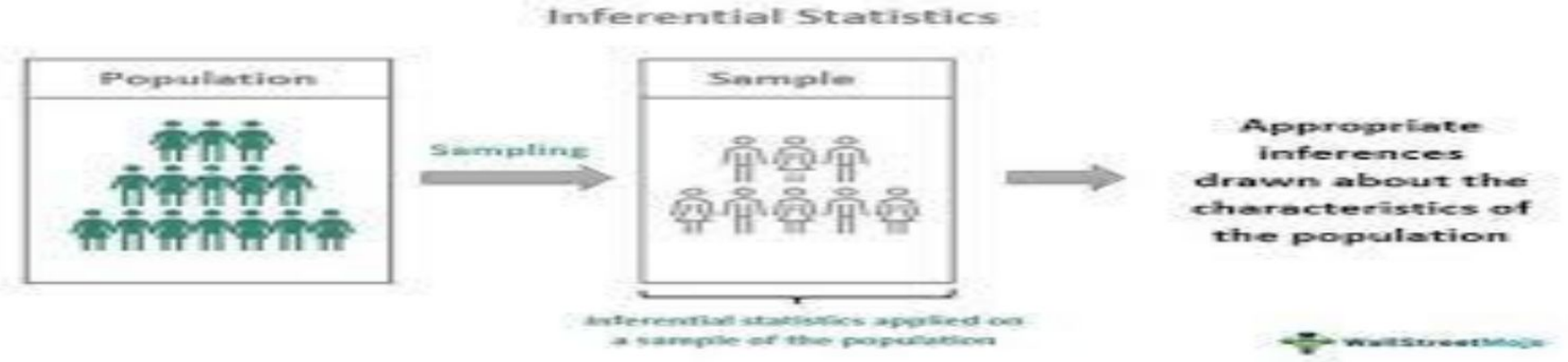
Inferential Statistics

Inferential statistics is a technique used to draw conclusions about a population by testing the data taken from the sample of that population.

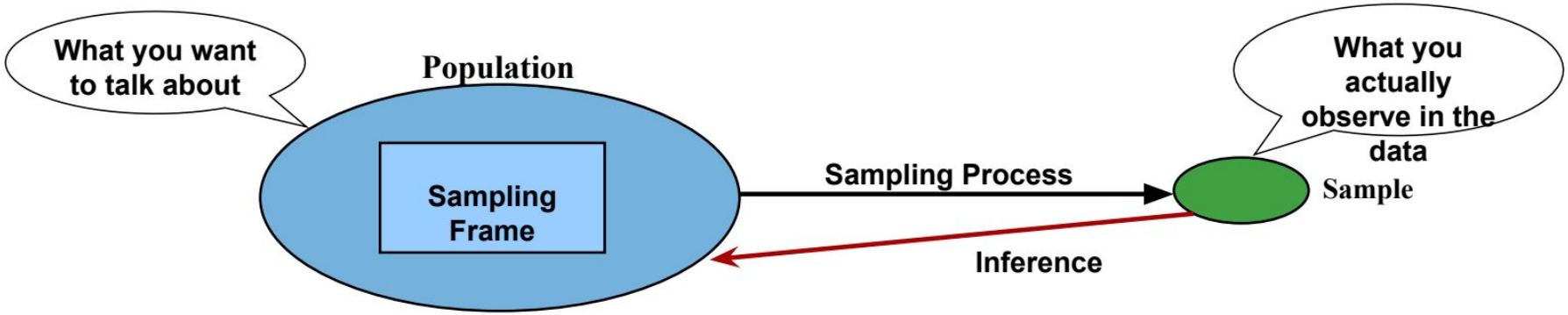
It is the process of how generalization from sample to population can be made. It is assumed that the characteristics of a sample is similar to the population's characteristics.

It includes testing hypothesis and deriving estimates.

It focuses on making statements about the population.



Process of Inferential Statistics



Raw Data

- It comprises of all the data collected from the sample.
- Depending on the sample size, this data can be large or small set of measurements.

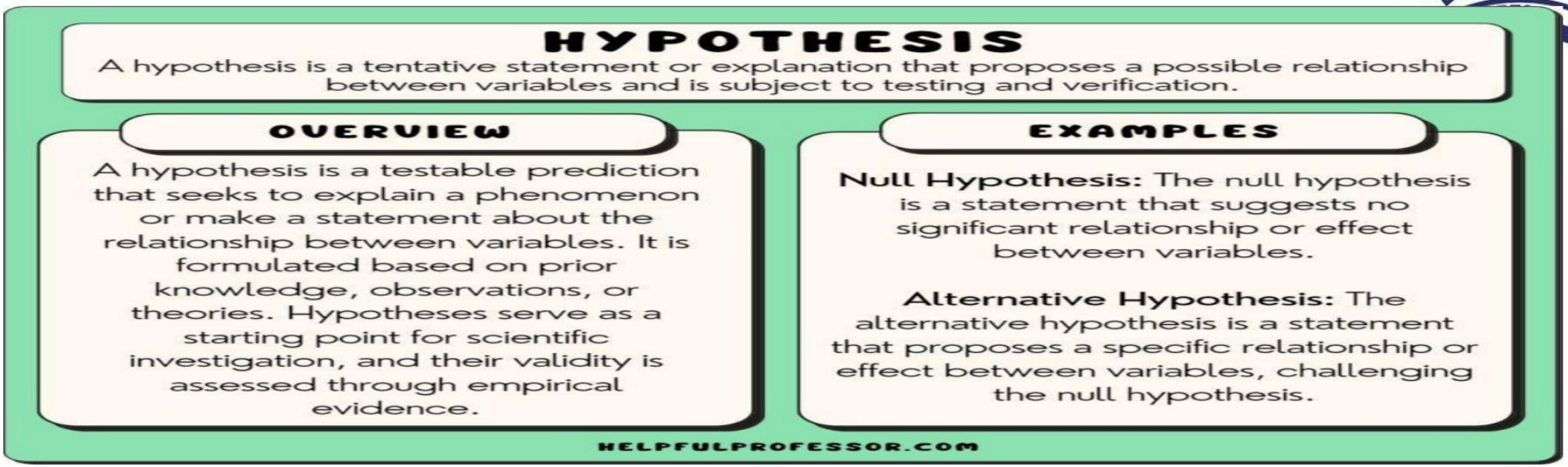
Sample Statistics

- It summarizes the raw data gathered from the sample of population
- These are the descriptive statistics (e.g. measures of central tendency)

Inferential Statistics

- These statistics then generate conclusions about the population based on the sample statistics.

Hypothesis Testing



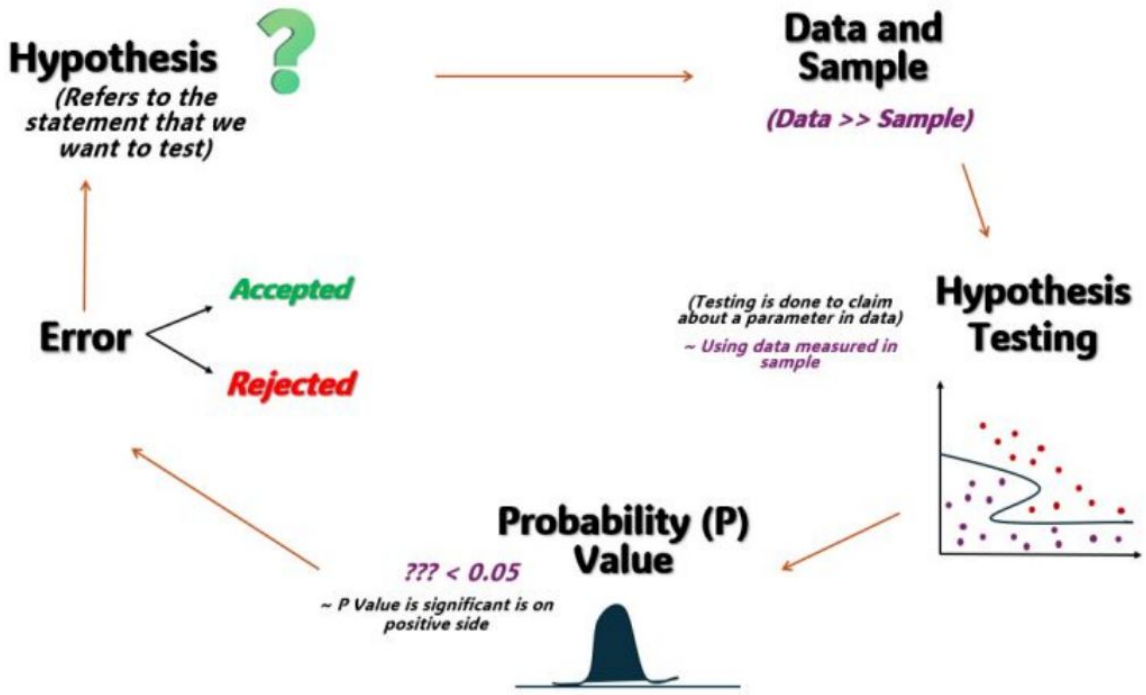
A research study and its hypotheses generally examine the relationships between independent and dependent variables – so you need to know these two concepts:

- The independent variable is the variable that is **causing** a change.
- The dependent variable is the variable the is **affected** by the change. This is the variable being tested.

Hypothesis Testing

Hypthesis Testing

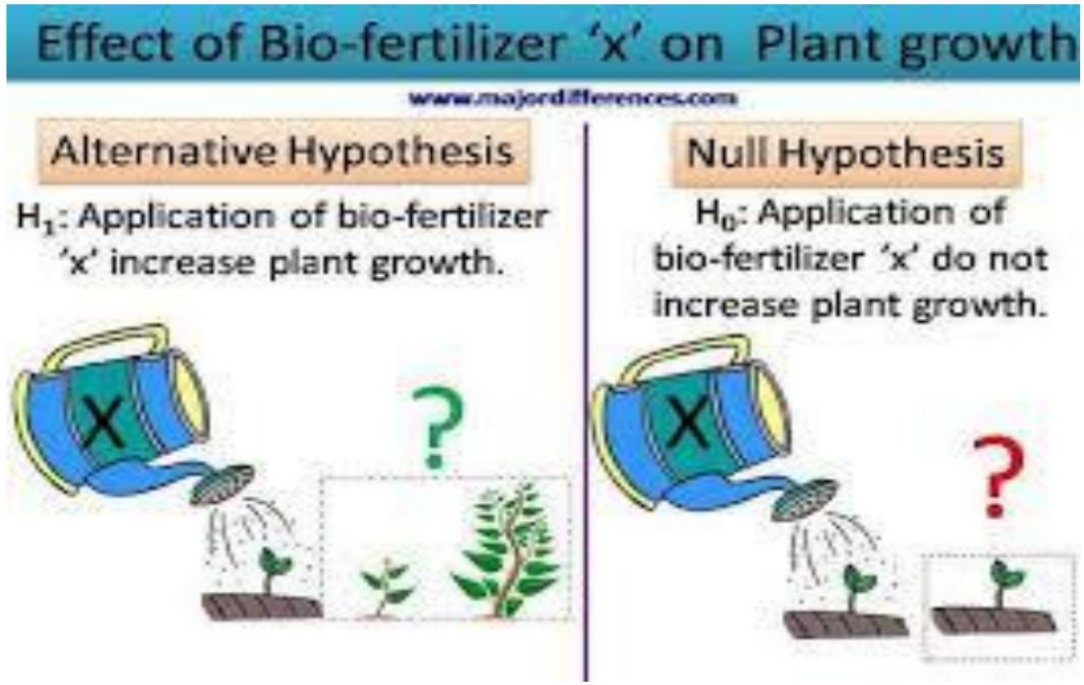
- It's like a detective game where we have two guesses about something in a group: one saying there's no difference, and the other saying there is.
- We collect data from a smaller group and use statistics to see if we can prove one guess is more likely.
- It helps us decide if our ideas about the whole group are true or not.



Hypothesis Testing (Null Vs Alternate)

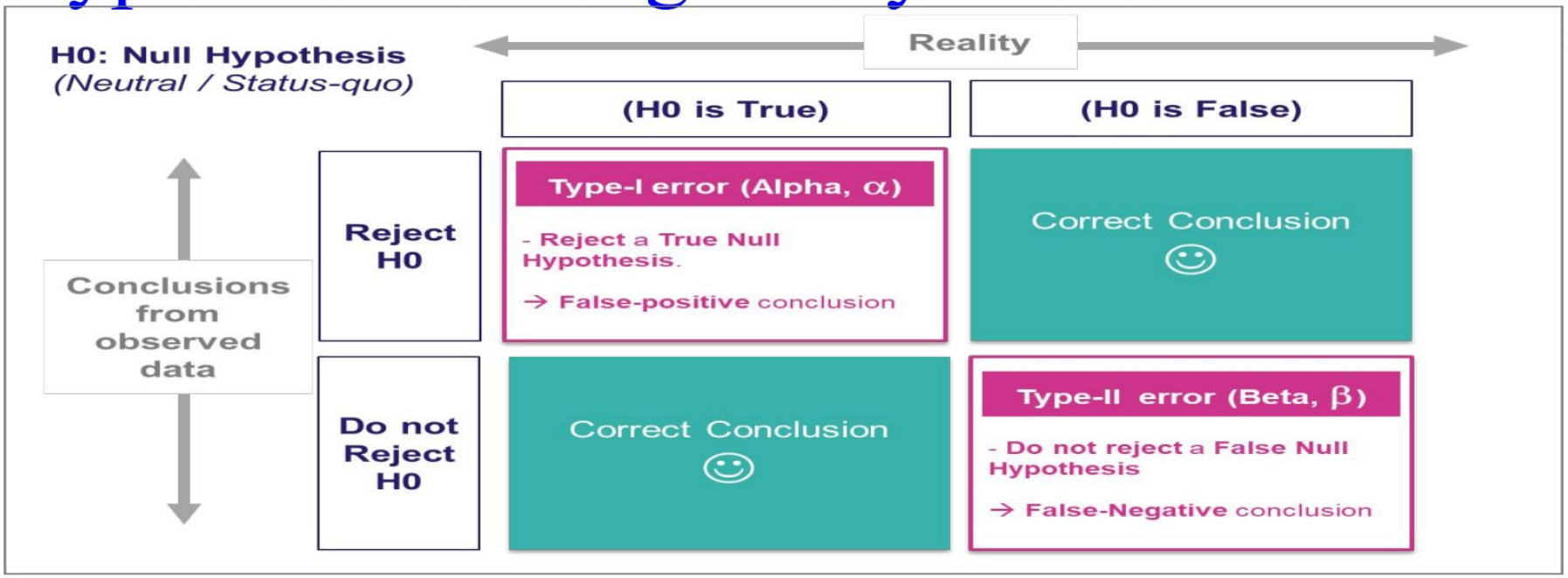
- **Null hypothesis**. The statement that there is no relationship between variables or no difference in the populations being tested. The null hypothesis is often written as H0.

- **Alternative hypothesis**. The statement that there is a relationship between variables or a difference in the populations being tested. The alternative hypothesis is often written as H1.



For Example: If one plant is fed with club soda for one month and another plantfed with plain water, there will be no difference in growth between the two plants.

Hypothesis Testing – Key Terms



Type I error: When we reject the null hypothesis, although that hypothesis was true. Type I error is denoted by alpha(αα).

Type II errors: When we accept the null hypothesis, but it is false. Type II errors are denoted by beta(ββ).

Hypothesis Testing – Key Terms



Significance/Confidence Level

- The significance level is used to decide whether the null hypothesis should be rejected or not.
- If the p-value is smaller than the significance level, the null hypothesis is to be rejected; otherwise, it is not to be rejected.

Critical value: The critical value in statistics is a threshold or cutoff point used to determine whether to reject the null hypothesis in a hypothesis test.

Probability (P) Value: The p-value helps us decide. If the p-value < 0.05 (commonly accepted threshold), the result is statistically significant. This means there is less than a 5% chance that the observed outcome occurred due to random variation

Hypothesis Testing – Z Test



Hypothesis Testing – Z Test

- Z-test is a statistical test that is used to determine whether the mean of a sample is significantly different from a known population mean when the population standard deviation is known.
- It is particularly useful when the sample size is large (>30).
- Z-test can also be defined as a statistical method that is used to determine whether the distribution of the test statistics can be approximated using the normal distribution or not.
- It is the method to determine whether two sample means are approximately the same or different when their variance is known and the sample size is large (should be >= 30).

Z- Test Formula

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \qquad Z = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\bar{x}$ is the mean of the sample, and μ is the assumed mean, σ is the standard deviation, and n is the sample size

One Tailed Z Test



One Tailed z-test

- One sample z-test is used to determine whether a particular population parameter, which is mostly mean, significantly different from an assumed value.
- A one-tailed test would be used when the study has to test whether the population parameter being tested is either lower than or higher than some hypothesized value.
- A one-sample z-test assumes that data are a random sample collected from a normally distributed population that all have the same mean and same variance.

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

where $\bar{x}$ is the mean of the sample, and **μ** is the assumed mean, **σ** is the standard deviation, and **n** is the sample size

Two Tailed Z Test



Two Tailed z-test

- In the case of two sample z-test, two normally distributed independent samples are required.
- A two-tailed z-test is performed to determine the relationship between the population parameters of the two samples.
- In the case of the two-tailed z-test, the alternative hypothesis is accepted as long as the population parameter is not equal to the assumed value.

$$z = \frac{\overline{X1} - \overline{X2}}{\sqrt{\sigma^2\left(\frac{1}{n1} + \frac{1}{n2}\right)}}$$

where $\bar{x}$ is the mean of the sample, and μ is the assumed mean, σ is the standard deviation, and n is the sample size

Z Test – example - 1



An ambulance service claims that it takes on average 8.9 minutes to reach its destination in emergency calls. To check the claim the agency which licenses ambulance services has then timed on 50 emergency calls, getting a mean of 9.3 minutes with a std deviation of 1.8 minutes. Does this constitute evidence that the figure claimed is too low at the 1% significant level?

Solution

Let us consider the null hypothesis Ho that the claim is same as observed

Ho:$\mu=8.9$, H1:$\mu = 8.9$

Given $n=50, x=9.3, s=1.8$

$Z=(x-\mu)/s_{\bar{X}} = x-\mu/(s/\sqrt{n}) =$

$(9.3-8.9)/(1.8/\sqrt{50})=1.574$

Since z(cal) is less than critical value $z(\alpha/2)= \pm 2.58$ the null hypothesis is accepted

Z Test – example - 2



A hospital uses large quantities of packaged doses of a particular drug. The individual dose of this drug is 100 cc. The action of this drug is such that body will harmlessly pass off excessive doses. On other hand insufficient doses do not produce desired medical effect. The hospital has purchase the drug from the same manufacturer for years and population standard deviation is 2 cc. The hospital inspects 50 doses of this drug at random and finds a mean of 99.75 cc

If a hospital sets a significance level of 0.10 and asks whether the dosage is too low how can we find this.

Solution

Let us consider the null hypothesis Ho that the claim is same as observed

Ho:$\mu o = 100$: H1 : $\mu 1 = 100$ (The hospital wishes to know whether actual dosage are 100cc or whether in fact the dosage are too small)

Given n=50,x=99.75 ,s=2

$Z=(x-\mu)/sX = x-\mu/(s/\sqrt{n}) =$

$(99.75-100)/(2/\sqrt{50})=0.88$

The critical value of z is -1.28 and therefore we accept null hypothesis

Z Test – example - 3



- A packaging device is set to fill detergent powder packets with a mean weight of 5 kg, with a standard deviation of 0.21 kg. The weight of packets can be assumed to be normally distributed.
- The weight of packets is known to drift upwards over a period of time due to machine fault which is not tolerable. A sample of 100 packets is taken and weighed. This sample has a mean weight of 5.03 kg. can we conclude that the mean weight produced by the machine has increased? Use a 5 % level of significance

Solution

- Let null hypothesis Ho that mean weight has increased
- Ho : $\mu \geq 5$ and H1 : $\mu < 5$
- Given n=100, x=5.03 kg, σ=0.21, α= 5%
- $Z = (x - \mu) / (\sigma/\sqrt{n}) = 5.03-5 /(0.21/\sqrt{100}) = 1.428$
- Since calculated value z(cal) = 1.428 is less than its critical value $z(\alpha) = 1.645$ at α = 0.05, the null hypothesis Ho is accepted
- Hence we conclude that mean weight is likely to be more than 5 kg

Z Test – example - 4



- A firm believes that the tyres produced by process A on an average last longer than tyres produced by process B.
- To test this belief, random supplies of tyres produced by the two processes were tested and the result are

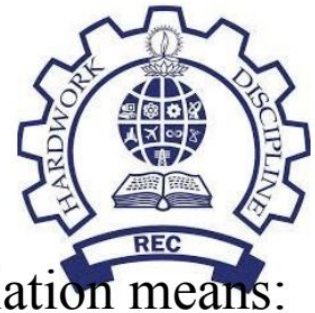
Process	Sample size	Average lifetime(km)	Standard deviation(km)
A	50	22400	1000
B	50	21800	1000

- Is there evidence at a 5 % level of significance that the firm is correct in its belief?

Solution

- Let us take the null hypothesis that there is no significant difference in the average life of tyres produced by processes A and B
- Ho : $\mu1=\mu2$ or $\mu1-\mu2 = 0$
- H1 : $\mu1=\mu2$
- Given x1 = 22400, x2 =21800
- $\sigma1 =\sigma2=1000$, n1=n2=50
- $Z= [(x1-x2) – (\mu1-\mu2)]/ \sigma x1 –x2$
- $= [(x1-x2) – (\mu1-\mu2)]/(\sqrt{\sigma_1^2/n1 + \sigma_2^2/n2})$

Z Tests – Results and Importance



Z Test – Interpreting the Results

- Z-score: Measures the number of standard deviations the sample mean is from the population mean.
- P-value: Represents the probability of observing a z-score as extreme or more extreme, assuming the null hypothesis (no difference between sample and population means) is true.
- If the p-value is less than a chosen significance level (e.g., 0.05), we reject the null hypothesis and conclude that there is a statistically significant difference between the sample mean and the population mean.

Z Test – Importance

- Comparing sample and population means: The Z-test is essential for determining if a sample mean is significantly different from a known population mean.
- Hypothesis testing: It's used to test hypotheses about population means.
- Statistical inference: The Z-test allows us to draw conclusions about a population based on sample data.

Hypothesis Testing – t Test (different types



There are three main types of t test:

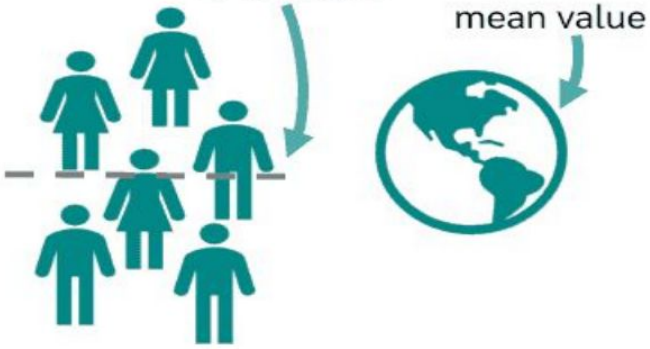
1. Independent Samples t-test compares the means for two groups.
2. Paired sample t-test or 2 sample t-test compares means from the same group at different times (say, one year apart).
3. One sample t-test tests the mean of a single group against a known mean.

t Test – One Sample Test

A manufacturer of chocolate bars claims that its chocolate bars weigh 50 grams on average. To verify this, a sample of 30 bars is taken and weighed. The mean value of this sample is 48 grams. We can now perform a one sample t-test to see if the mean of 48 grams is significantly different from the claimed 50 grams.

1.

One sample t-test



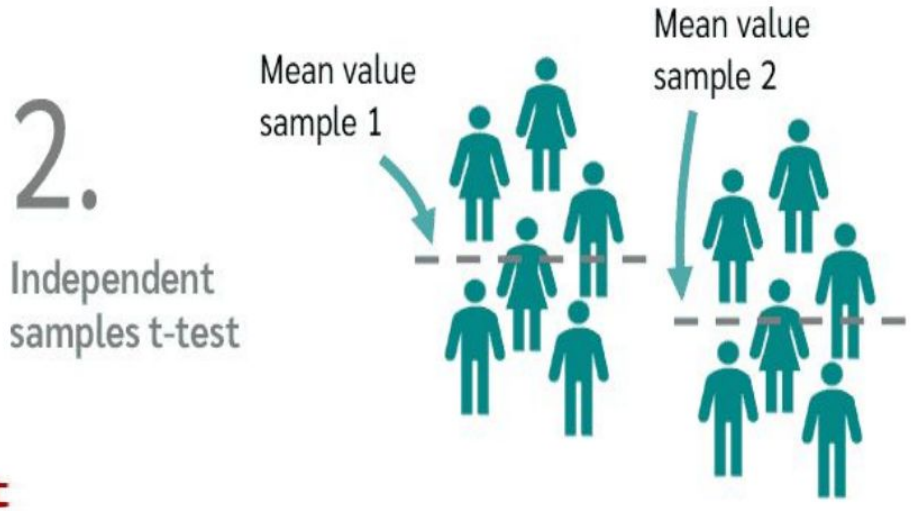
One-Sample T-Test

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$\bar{x}$ = obersved mean of the sample
μ = assumed mean
s = standard deviation
n = sample size

t Test – Two Sample Test

When to use the t-test for independent samples? We use the t-test for independent samples when we want to compare the means of two independent groups or samples. We want to know if there is a significant difference between these means.



Two-Sample T-Test

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

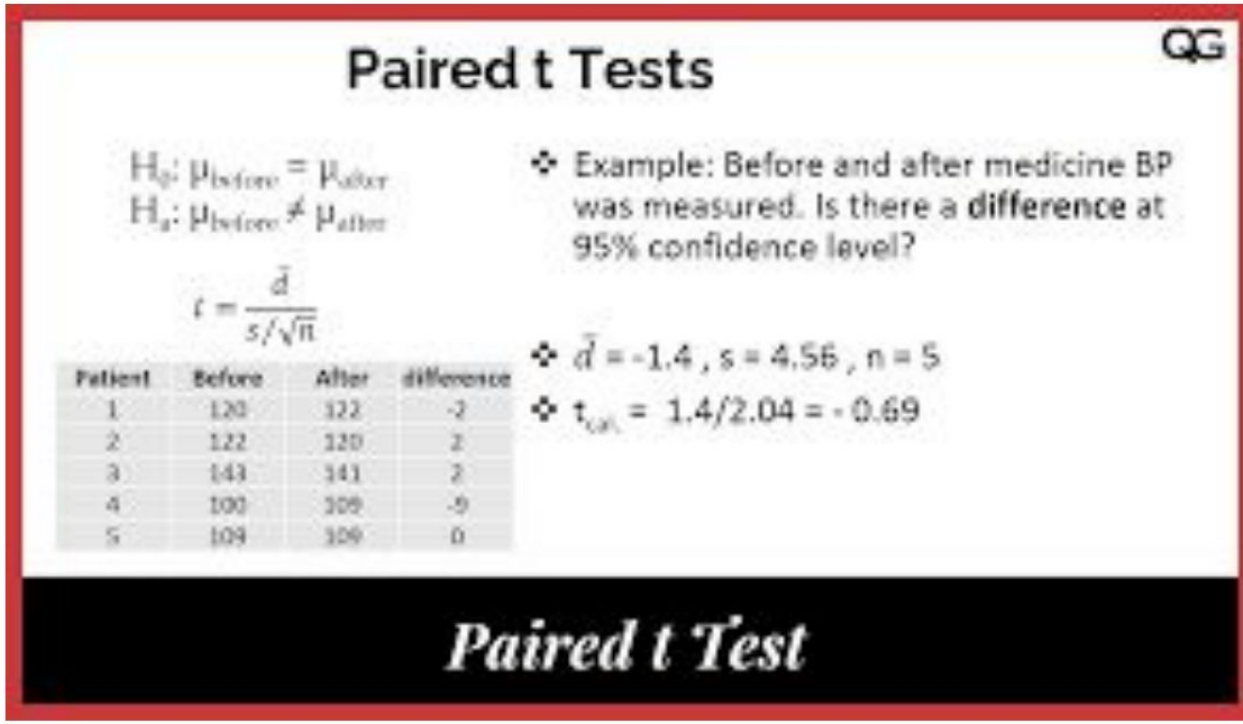
$\bar{x}_1$ = observed mean of 1st sample
$\bar{x}_2$ = observed mean of 2nd sample
s_1 = standard deviation of 1st sample
s_2 = standard deviation of 2nd sample
n_1 = sample size of 1st sample
n_2 = sample size of 2nd sample

Hypothesis Testing – Paired T Test

Hypothesis Testing – T Test

Paired sample t-test, commonly known as dependent sample t-test is used to find out if the difference in the mean of two samples is 0.

The test is done on dependent samples, usually focusing on a particular group of people or things. In this, each entity is measured twice, resulting in a pair of observations.



T Test – example - 1



Example:

A institute claims that, they pay more than Rs 80000 salary to their employees. To check this claim salary of 10 employees are as follows—
84250,65800,68900,78900,62000,59870,72900,80200,79350,71260
Test whether claim is true.

H_0 = Mean salary paid to employees is Rs 80000

H_1 = Mean salary paid to employees is ≥Rs 80000

Given: μ=Rs 80000, $\bar{x} = \frac{\sum X}{n} = Rs72343$, s= $\sqrt{\frac{\sum(X-\bar{X})^2}{n}}$ =8261.71, n=10

and t=-2.79

Critical value of t with n-1=9 degrees of freedom at 5% level of significance=2.262

$|-2.79|=2.79>2.262$, Reject H_0

Conclusion: Mean salary paid to employees is ≥Rs 80000

T Test – example - 2



Example:

Running capacity of two horses (in hours) on track for 8 days given as follows-Horse A: 10,12,8,9,10,11,9,10 and
Horse B: 12,10,9,10,11,12,10,11

Is two horses have same running capacity.
$H_0 =$ Two horses have same running capacity
$H_1 =$ Two horses have different running capacity

Given: $\bar{X}_1 = \frac{\sum X_1}{n_1} = 9.875,$
$\bar{X}_2 = \frac{\sum X_2}{n_2} = 10.625, S^2 = \frac{\sum (X_1 - \bar{X}_1)^2 + \sum (X_2 - \bar{X}_2)^2}{n_1 + n_2 - 2} = 1.5305,$
and t= 1.21
Critical value of t with $n_1 + n_2 - 2$=14 degrees of freedom at 5% level of significance=2.145
1.21<2.145, Accept H_0
Conclusion: Two horses have same running capacity

T Tests – Interpreting the Results



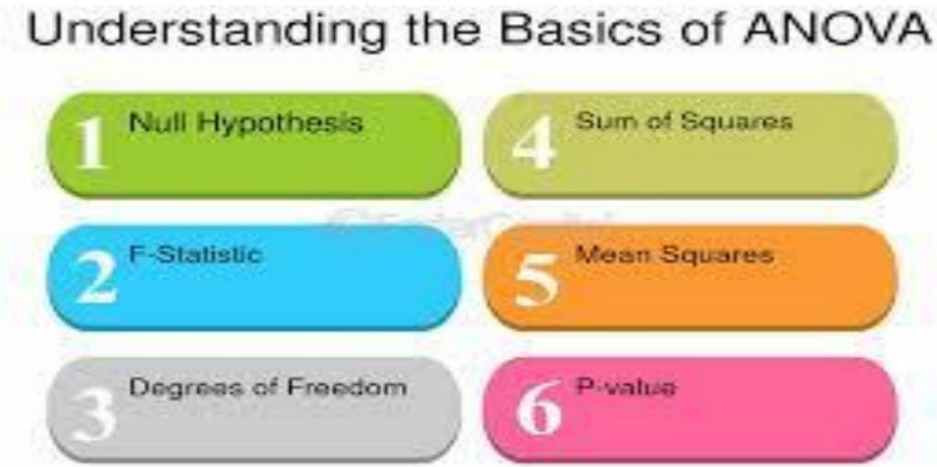
T Test – Interpreting the Results

- T-statistic: Measures the difference between the groups means relative to the variability within the groups.
- P-value: Represents the probability of observing a t-statistic as extreme or more extreme, assuming the null hypothesis (no difference between means) is true.
- If the p-value is less than a chosen significance level (e.g., 0.05), we reject the null hypothesis and conclude that there is a statistically significant difference between the means of the two groups.

T Test – Importance

- Hypothesis testing: T-tests are crucial for comparing means and making informed decisions in various fields, including research, medicine, and social sciences.
- Statistical inference: They allow us to draw conclusions about a population based on sample data.
- Decision-making: T-tests can help researchers determine the effectiveness of treatments, interventions, or experimental conditions.

Hypothesis Testing – **ANOVA** (Analysis of Variance)



Hypothesis Testing – ANOVA

- ANOVA, or Analysis of Variance is a parametric statistical technique that helps in finding out if there is a significant difference between the mean of three or more groups.
- ANOVA works by comparing the variance between groups to the variance within each group.

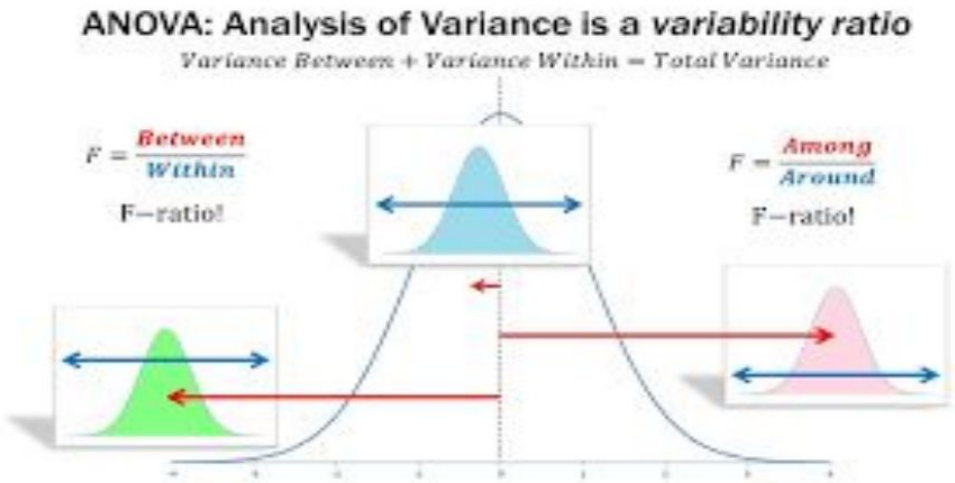
How Does ANOVA Work?
Variance Between Groups: This measures how much the group means differ from the overall mean.
Variance Within Groups: This measures how much the data points within each group differ from their group mean.

Hypothesis Testing – **ANOVA** (Analysis of Variance)

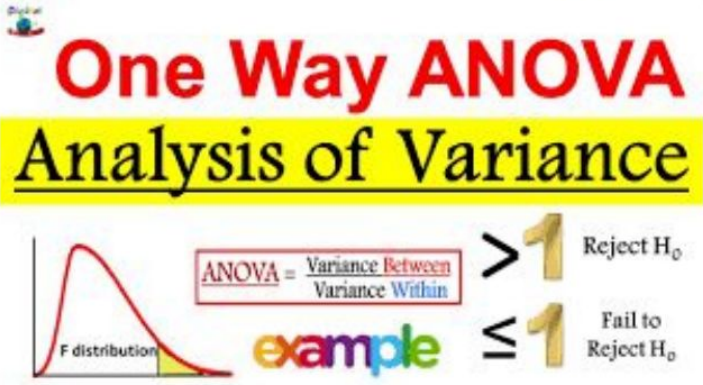
Hypothesis Testing – ANOVA

ANOVA calculates a value called the F-ratio by dividing the variance between the groups by the variance within the groups.

If the F-ratio is significantly large, it suggests that the differences between the groups are larger than would be expected by chance alone.



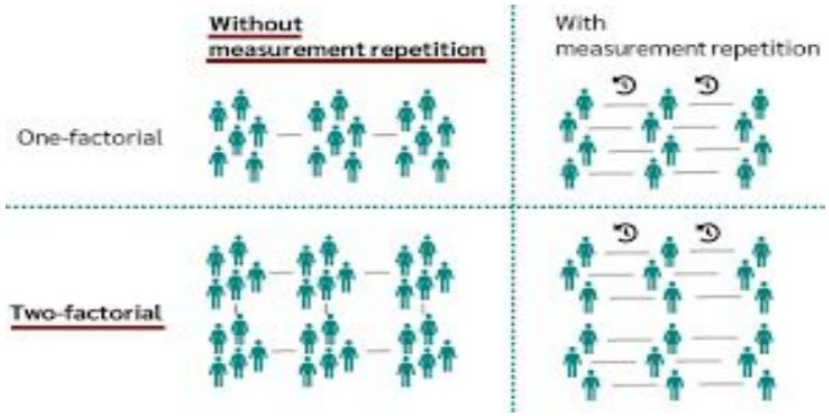
Hypothesis Testing – **ANOVA** (Analysis of Variance)



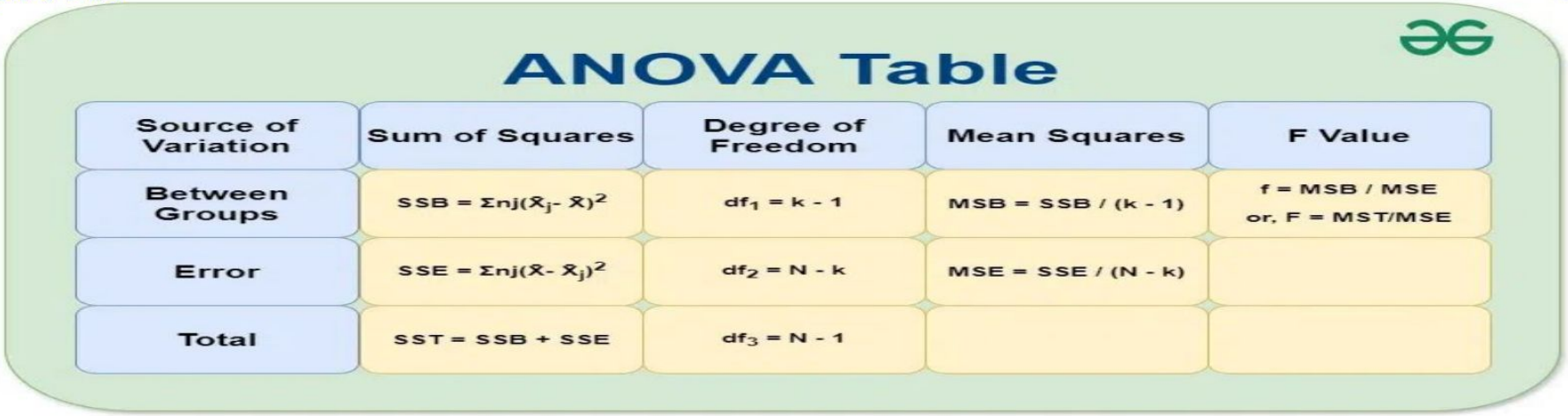
Types of ANOVA

1. **One-way ANOVA**: This is used when comparing the average of three or more groups based on one factor. For example, comparing the average test scores of students from three different schools.

2. **Two-way ANOVA**: Two-way ANOVA involves two independent variables. It allows for the examination of the main effects of each variable as well as the interaction between them.
   For example, looking at the effect of different fertilizers and different watering frequencies on plant growth.



Hypothesis Testing – Elements of Anova Table



ANOVA Table

Source of Variation	Sum of Squares	Degree of Freedom	Mean Squares	F Value
Between Groups	$SSB = \Sigma nj(\bar{x}_j - \bar{x})^2$	$df_1 = k - 1$	$MSB = SSB / (k - 1)$	$f = MSB / MSE$ or, $F = MST/MSE$
Error	$SSE = \Sigma nj(\bar{x} - \bar{x}_j)^2$	$df_2 = N - k$	$MSE = SSE / (N - k)$	
Total	$SST = SSB + SSE$	$df_3 = N - 1$		

where,

- F = ANOVA Coefficient
- MSB = Mean of the total of squares between groupings
- MSW = Mean total of squares within groupings
- MSE = Mean sum of squares due to error
- SST = total Sum of squares
- p = Total number of populations
- n = The total number of samples in a population
- SSW = Sum of squares within the groups
- SSB = Sum of squares between the groups
- SSE = Sum of squares due to error
- s = Standard deviation of the samples
- N = Total number of observations

Hypothesis Testing – Anova Example



Example -1

Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significance.

Plots of land	per acre production data		
	variety of wheat		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

Hypothesis Testing – Anova Example



Step 1 : obtain the mean of each sample i.e., obtain

$$\bar{X}_1 = \frac{6+7+3+8}{4} = 24/4 = 6$$

$$\bar{X}_2 = \frac{5+5+3+7}{4} = 20/4 = 5$$

$$\bar{X}3 = \frac{5+4+3+4}{4} = 16/4 = 4$$

Hypothesis Testing – Anova Example



Step 2: Find mean of sample mean

Mean of the sample mean or $\bar{\bar{X}} = \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3}{k}$

$$= \frac{6+5+4}{3}$$

$$= 5$$

Step 3: Now we work out between and SS within samples:

$$\text{SS between} = n_1 (\bar{x}_1 - \bar{\bar{x}})^{2+} n_2 (\bar{x}_2 - \bar{\bar{x}})^2 + n_3(\bar{x}_3 - \bar{\bar{x}})^2$$
$$= 4(6-5)^2 + 4(5-5)^2 + 4(4-5)^2$$
$$= 4+0+4$$
$$= 8$$

Hypothesis Testing – Anova Example



Step : calculate the SS within the Sample

$$\text{SS within} = \sum(X_{1i}-\bar{\bar{X}}_1)^2 + \sum(X_{2i}-\bar{\bar{X}}_2)^2 + \sum(X_{3i}-\bar{\bar{X}}_3)^2$$

$$=(6-6)^2 + (7-6)^2 + (3-6)^2 + (8-6)^2$$

$$+(5-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2 +$$

$$+(5-4)^2 + (4-4)^{2+} (3-4)^2 + (4-4)^2$$

$$= (0+1+9+4) + (0+0+4+4) + (1+0+1+0)$$
$$= 14+8+2$$
$$= 24$$

Hypothesis Testing – Anova Example



- Now calculate total variance

SS for total variance $= \sum (X_{ij} - \bar{\bar{X}})^2$

$= (6-5)^2 + (7-5)^2 + (3-5)^{2+} (8-5)^2$
$+ (5-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2$
$+ (5-5)^2 + (4-5)^2 + (3-5)^2 + (4-5)^2$

$= 1+4+4+9+0+0+4+4+0+1+4+1$
$=32$

- Alternatively it (SS for total variance) can also be worked out thus:

SS for total = SS between + SS within
= 8 + 24
= 32

Hypothesis Testing – Anova Example

Table : one way anova

Source of variation	SS	d.f.	MS	f- ratio	5% F-limit(from the f ratio table)
Between sample	8	(3-1)= 2	8/2 = 4	4/2.67= 1.5	F(2,9)= 4.26
Within Sample	24	(12-3) = 9	24/9 = 2.67		
Total	32	(12-1) = 11			

Hypothesis Testing – Anova Example

- The above table shows that the calculated value of *F is 1.5 which is less than the table value of 4.26 at 5% level with* d.f. being *v1 = 2 and v2 = 9 and hence could have arisen due to chance.*
- *This analysis supports the null-hypothesis of* no difference is sample means.
- The difference in wheat output due to varieties is insignificant and is just a matter of chance.
- **It means that the varieties is not so much differ from each other**

Hypothesis Testing – Anova Example - 2

Example -2

Set up an analysis of variance table for the following two-way design results:

	per acre production data		
	Varieties of seeds		
Varieties of fertilizers	A	B	C
W	6	5	5
X	7	5	4
Y	3	3	3
Z	8	7	4

Hypothesis Testing – Anova Example - 2



Step1 : Obtain the total value of individual

Total = sum of all individual
= 6+7+3+8+5+5+3+7+5+4+3+4
=60

so we have T= 60 and n = 12

Where
T = sum of all individual sample
N = number of sample

Step 2 : calculate correction factor

Correction factor = Square of total value / number of sample

So correction factor = $(T)^2$ /n

= 60*60/12 = 300

Hypothesis Testing – Anova Example - 2



Step 2 : calculate Total SS

Total SS = square of all the items value – correction factor

= {(36+25+25+49+25+16+9+9+9+64+49+16) – (60*60/12)}

= 332-300

= 32

So we have total SS = 32

Individual value	squaring	Result of squaring
6	6*6	36
5	5*5	25
5	5*5	25
7	7*7	49
5	5*5	25
4	4*4	16
3	3*3	9
3	3*3	9
3	3*3	9
8	8*8	64
7	7*7	49
4	4*4	16

Hypothesis Testing – Anova Example - 2



Step 3 : calculate SS between columns treatment

SS between column = sum of all (sqaure of individual column / sample in column) – correction factor

= (24*24/4 + 20*20/4+16*16/4) – (60*60/12)

= 144 +100+60 – 300

=8

24*24	576/4 = 144
20*20	400/4= 100
16*16	256/4=60

Hypothesis Testing – Anova Example - 2

Step 4 : calulate SS Between rows treatment

SS between = sum of all square of individual row / number of sample) – correction factor

= (16*16/3 + 16*16/3 +9*9/3)
—(60*60/12)
= 85.33 + 85.33 +27+120.33 – 300
= 317.99 - 300
=18

16*16	256/3=85 .33
16*16	256/3=85 .33
9*9	81/3=27
19*19	361/3 = 120.33

Hypothesis Testing – Anova Example - 2

Step 5 : SS residual or error

SS residual = Total SS – (SS between columns + SS between rows)
=32 - (8+18)
= 6

Step 6 : find degree of freedom

d.f. for variance between columns = (c – 1)
= 3-1 = 2

d.f. for variance between rows = (r – 1)
= 4-1 = 3

d.f. for residual variance = (c – 1) (r – 1)
= 2*3 = 6

d.f. for total variance = (c . r – 1)
= 3*4 - 1
= 12 - 1 = 11

Hypothesis Testing – Anova Example - 2

Finally we have some important quantitative value

SS between column	8
SS between row	18
SS residual	6
d.f for variation between column	2
d.f for variation between row	3
d.f for residual variance	6
d.f for total variance	11

Hypothesis Testing – Anova Example - 2

Table : Two way Anova Table

Sample of variation	Sum of square (SS)	Degree of freedom (d.f.)	Mean square(MS)	F –ratio	5% F – limit (or the tables values)
Between colums treatment	8	(3-1) = 2	8/2 = 4	4/1 = 4	F (2,6) = 5.14
Between rows treatment	18	(4-1) = 3	18/3 = 6	6/1 = 6	F (3,6)= 4.76
Residual or Error	6	(3-1)*(4-1) = 6	6/6 = 1		
Total	32	(3*4)-1			

Hypothesis Testing – Anova Example - 2

Conclusion

The above table shows that the calculated value of *F is 4 which is less than the table value 5.14 of at 5% level with* d.f. being *v1 = 2 and v2 = 6 in column .*

a) This analysis show has no significance .

The f value in rows is 6 which is higher than the table 4.76 of a 5% level with d.f. Being v1=3 and v2= 6.

***b)This analysis supports the null-hypothesis which shows* difference is sample means.**

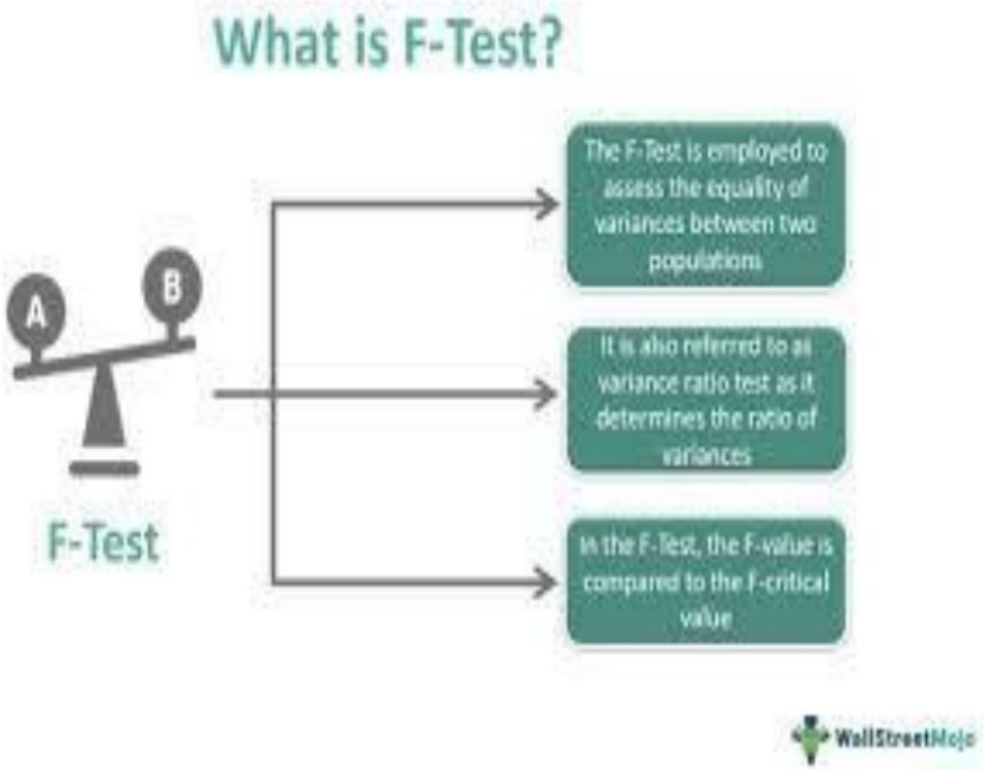
Hypothesis Testing – Anova Example - 2

- It means that differences concerning varieties of seeds are insignificant at 5% level as the calculated F ratio of 4 is less than the table value of 5.14 but the variety differences concerning fertilizers are significant as the calculated F ratio of 6 is more than its table value of 4.76
- Fertilizers act differently, the different varieties of seeds are affected differently.

Hypothesis Testing – F Test

Hypothesis Testing – F Test

- In statistical hypothesis testing, the Fisher-Snedecor distribution, sometimes known as the F-distribution, is used, particularly when comparing variances or



Hypothesis Testing – F Test

$F = \frac{S_1^2}{S_2^2}$

S_1^2 = Estimate of Variance of sample1 = $\frac{\sum(X_1 - X_1)^2}{n_1 - 1}$

S_2^2 = Estimate of Variance of sample1 = $\frac{\sum(X_2 - X_2)^2}{n_2 - 1}$

Degrees of freedom v1 = $n_1 - 1$; v2= $n_2 - 1$

Hypothesis Testing – F Test Assumtions

- Normality – Data are normally distributed
- Homogeneity – variance within each group are equal
- Independence of error –the error is independent for each group

Hypothesis Testing – F Test - example

In a sample of 8 observations, the sum of squared deviations of items from the mean was 84.4. In another samples of 10 observations, the value was found to be 102.6. Test whether the difference is significant at 5% level.

Solution

Let us take hypothesis that the difference in the variance of two samples is not significant $H_0: \sigma_1^2 = \sigma_2^2$

n1= 8 $\sum(X_1 - X_1)^2$= 84.4 n2 = 10 $\sum(X_2 - X_2)^2$= 102.6

$$S_1^2 = \frac{\sum(X_1 - X_1)^2}{n_1 - 1} = \frac{84.4}{8-1} = \frac{84.4}{7} = 12.06$$

$$S_2^2 = \frac{\sum(X_2 - X_2)^2}{n_1 - 1} = \frac{102.6}{10-1} = \frac{102.6}{9} = 11.4$$

$$F = \frac{S_1^2}{S_2^2} = \frac{12.6}{11.4} = 1.06$$

Calculated value of F = 1.06
Table value of F = 3.29 (v1 = 7 and v2 = 9 at 5% level of significance)
Inference
CV < TV, hence we accept the null hypothesis and conclude that the difference in the variance of two samples is not significant at 5%

V1 = 7 & V2 = 9 degrees of freedom at 5% level of significance $F_{0.05}$ = 3.29

F-TABLE: VALUES OF F OF THE F-DISTRIBUTION (ALPHA = 0.05)



DEGREE OF NUMERATOR (v1)

v2	1	2	3	4	5	6	7	8	9	10
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16

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